EXERCISE:01

1. **OUTPUT:**

```
> roll_no
[1] 315
> name
[1] "Divya"
> cgpa
[1] 9.12
> pass
[1] FALSE
>
```

2. **OUTPUT:**

```
> emp_id
[1] 7420
> salary
[1] 38999.25
> department
[1] "Marketing"
> is_permanent
[1] TRUE
>
```

3. **OUTPUT:**

```
> product_name
[1] "Headphones"
> price
[1] 2499
> stock_available
[1] 60
> discount_eligible
[1] FALSE
```

4. **OUTPUT:**

```
> city
[1] "Hyderabad"
> temperature
[1] 28.7
> rain
[1] TRUE
> wind_speed
[1] 6.3
>
```

5. **OUTPUT:**

```
> subject
[1] "Chemistry"
> total_marks
[1] 150
> marks_obtained
[1] 112
> result
[1] TRUE
```

6. **OUTPUT:**

```
> account_no
[1] 556677889
> holder_name
[1] "Arjun Mehta"
> balance
[1] 10500
> account_active
[1] FALSE
```

7. **OUTPUT:**

```
> movie_title
[1] "Avatar"
> rating
[1] 7.9
> reviews
[1] 875
> available
[1] TRUE
```

8. **OUTPUT:**

```
> vehicle_count
[1] 210
> location
[1] "Ring Road"
> avg_speed
[1] 55.2
> congested
[1] FALSE
```

EXERCISE:02

1. **OUTPUT:**

```
[1] "Not Eligible for Loan"
```

2. **OUTPUT:**

```
[1] "Not Eligible for Exam"
```

3. **OUTPUT:**

```
[1] "Discount Applied"
```

4. **OUTPUT:**

```
[1] "Low Usage"
```

5. **OUTPUT:**

```
[1] "Not Eligible for Bonus"
```

6. **OUTPUT:**

```
[1] "Invalid Password"
```

7. **OUTPUT:**

```
[1] "Extreme Temperature Warning"
```

8. **OUTPUT:**

```
[1] "Fail"
```

9. **OUTPUT:**

```
[1] "Credit Card Rejected"
```

10. **OUTPUT:**

```
[1] "Medium Risk"
```

11. **OUTPUT:**

```
[1] "Short Green Signal"
```

12. **OUTPUT:**

```
[1] "Not Eligible for Appraisal"
```

13. **OUTPUT:**

```
[1] "Exam Disqualified"
```

14. **OUTPUT:**

```
[1] "Normal Pricing"
```

15. **OUTPUT:**

```
[1] "Machine Shutdown"
```

16. **OUTPUT:**

```
[1] "Admission Rejected"
```

EXERCISE:03

1. **OUTPUT:**

```
[1] 2700
```

2. **OUTPUT:**

```
[1] "Fail"
```

3. **OUTPUT:**

```
[1] 40950
```

4. **OUTPUT:**

```
[1] 425
```

5. **OUTPUT:**

```
[1] 1800
```

6. **OUTPUT:**

```
[1] 71.6
```

7. **OUTPUT:**

```
[1] "Fail"
```

8. **OUTPUT:**

```
[1] "Eligible to Vote"
```

9. **OUTPUT:**

```
[1] 15279.18
```

10. **OUTPUT:**

```
[1] 8715
```

11. **OUTPUT:**

```
[1] 4680
```

12. **OUTPUT:**

```
[1] "Second Class"
```

13. **OUTPUT:**

```
[1] 900
```

14. **OUTPUT:**

```
[1] 7200
```

15. **OUTPUT:**

```
[1] 32604
```

16. **OUTPUT:**

```
[1] "High Usage"
```

EXERCISE:04

1. **OUTPUT:**

```
  RollNo   Name  Dept Marks
1    201   Asha  MECH    67
2    202 Vikram CIVIL    74
3    203 Neeraj   EEE    81
4    204  Pooja  MECH    79
```

2. **OUTPUT:**

```
  EmpID   Name    Dept Salary
1    11  Kunal   Admin  28000
2    12 Ritika Support  32000
3    13    Dev      IT  41000
4    14 Farhan      HR  36000
```

3. **OUTPUT:**

```
  PatientID   Name Age  Cost
1       301  Mohan  60 22000
2       302  Geeta  29  9000
3       303 Ramesh  47 16000
```

4. **OUTPUT:**

```
  ProductID ProductName Quantity
1        10    Notebook      200
2        20      Marker      120
3        30      Folder       75
4        40     Stapler       50
```

5. **OUTPUT:**

```
  RollNo  Name Dept Sem1 Sem2 Attendance
1     11 Irfan  BIO   65   70         85
2     12 Kavya CHEM   72   75         80
3     13 Rohit  PHY   88   90         92
4     14 Leena  BIO   78   82         88
5     15 Manoj  PHY   83   86         91
```

6. **OUTPUT:**

	EmpID	Name	Dept	Basic	HRA	DA	Rating
1	201	Sanjay	Sales	27000	5400	2700	3.9
2	202	Priya	IT	38000	7600	3800	4.6
3	203	Varun	Marketing	35000	7000	3500	4.3

7. **OUTPUT:**

	PatientID	Name	Age	Disease	Amount
1	401	Nisha	33	Asthma	8000
2	402	Ajay	58	Arthritis	14000
3	403	Kiran	46	Migraine	6000

8. **OUTPUT:**

	ProductID	ProductName	UnitsSold	Revenue
1	5	TV	45	900000
2	6	Refrigerator	30	750000
3	7	AC	25	650000
4	8	Washing Machine	20	400000

9. **OUTPUT:**

	FieldID	Crop	Moisture	Temp	ExpectedYield
1	10	Cotton	50	34	3.5
2	20	Sugarcane	42	30	5.2
3	30	Soybean	48	33	4.0

10. **OUTPUT:**

	RollNo	Name	Dept	Sem1	Sem2	Attendance
1	21	Tejas	Maths	55	60	78
2	22	Nikita	Stats	68	70	82
3	23	Harish	CS	77	80	88

11. **OUTPUT:**

	EmpID	Name	Dept	Basic	HRA	DA	Rating
1	301	Anil	IT	25000	5000	2500	4.1
2	302	Bhavna	HR	30000	6000	3000	NA
3	303	Chetan	Sales	NA	NA	NA	4.7

12. **OUTPUT:**

	PatientID	Name	Age	Disease	Amount
1	501	Deepak	40	Flu	5000
2	502	Shalini	35	<NA>	7000
3	503	Omkar	62	Cancer	20000

13. **OUTPUT:**

	ProductID	ProductName	UnitsSold	Revenue
1	11	Fan	60	300000
2	12	Cooler	40	NA
3	13	Heater	30	200000
4	14	Geyser	NA	150000

14. **OUTPUT:**

```
  FieldID   Crop Moisture Temp ExpectedYield
1     101 Barley       35   28           2.8
2     102 Millet       42   31            NA
3     103 Pulses       NA   NA           3.6
```

EXERCISE:05

1. **OUTPUT:**

```
> upper_names
[1] "KIRAN"  "SOMA"   "RAJESH"
> name_length
[1] 5 4 6
```

2. **OUTPUT:**

```
[1] "deepa" "varun" "latha"
```

3. **OUTPUT:**

```
[1] "Machine Learning" "Cloud Computing"  "Big Data"
```

4. **OUTPUT:**

```
[1]  TRUE FALSE  TRUE
```

5. **OUTPUT:**

```
[1] "issue found in module, critical issue detected"
```

6. **OUTPUT:**

```
> payroll
  EmpID   Name    Dept Basic  HRA   DA Rating
1   201 Deepak      IT 40000 8000 4000    4.7
2   202  Meera      HR 32000 6400 3200    4.0
3   203 Suresh Finance 45000 9000 4500    4.6
```

7. **output:**

```
> hospital_data
  PatientID    Name Age  Disease Amount
1       401 Lakshmi  36   Asthma   9000
2       402   Kumar  47    Fever   5000
3       403    Anil  55 Diabetes  20000
```

8. **output:**

```
> sales_data
  ProductID     ProductName UnitsSold Revenue
1         5              TV        40 2000000
2         6          Fridge        30 1500000
3         7 Washing Machine        25 1000000
4         8              AC        35 1400000
```

9. **output:**

```
> agri_data
  FieldID      Crop Moisture Temp ExpectedYield
1      11    Cotton       42   33           3.9
2      12 Sugarcane       36   30           4.2
3      13      Corn       39   31           4.0
```

10. **output:**

```
> final_records
  RollNo  Name Dept Sem1 Sem2 Attendance
1     10  Amit   IT   65   70         90
2     20 Pooja  CSE   78   80         85
3     30 Rahul  ECE   88   85         88
```

11. **output:**

```
> payroll
  EmpID  Name    Dept Basic  HRA   DA Rating
1     1  Arun      IT 28000 5600 2800    4.3
2     2 Divya      HR 36000 7200 3600     NA
3     3 Kiran Finance    NA   NA   NA    4.9
```

12. **output:**

```
> hospital_data
  PatientID  Name Age Disease Amount
1        11  Ravi  40   Fever   5000
2        12  Sita  35    <NA>   7000
3        13 Arjun  50   Heart  20000
```

13. **output:**

```
> sales_data
  ProductID ProductName UnitsSold Revenue
1         1       Phone        50   5e+05
2         2      Laptop        40   2e+06
3         3      Tablet        20      NA
4         4       Watch        NA   3e+05
```

14. **output:**

```
> agri_data
  FieldID  Crop Moisture Temp ExpectedYield
1       1  Rice       44   31           4.3
2       2 Maize       39   28            NA
3       3 Wheat       NA   NA           3.9
```

EXERCISE:05

1. **OUTPUT:**

```
> upper_names
[1] "KIRAN" "LATHA" "VIJAY"
> name_length
[1] 5 5 5
```

2. **OUTPUT:**

```
> usernames
[1] "kiran" "latha" "vijay"
```

3. **OUTPUT:**

```
> clean_input
[1] "Machine Learning" "Big Data"         "Cloud"
```

4. **OUTPUT:**

```
> valid_codes
[1]  TRUE FALSE  TRUE
```

5. **OUTPUT:**

```
> updated_report
[1] "issue found in module, issue occurred again"
```

6. **OUTPUT:**

```
[1] "Weak Password"
```

7. **OUTPUT:**

```
> word_c
[1] 6
```

8. **OUTPUT:**

```
> extensions
[1] "txt" "ppt" "jpg"
```

9. **OUTPUT:**

```
> abbr
[1] "Sow" "Pra" "Har"
```

10. **OUTPUT:**

```
> matched_names
[1] "Kiran" "Latha" "Mohan"
```

11. **OUTPUT:**

```
> clean_students
  Name          Email Department Username ValidEmail
1 KAVYA kavya@univ.edu         IT    kavya       TRUE
2 ROHIT rohit@univ.edu        CSE    rohit       TRUE
```

12. **OUTPUT:**

```
>
> products
    Product        Code ProductCode
1    Iphone PRD-501-IPH         IPH
2   Sony-Tv PRD-602-SON         SON
3 Hp Laptop PRD-703-HPL         HPL
```

13. **OUTPUT:**

```
> comments
  User             Comment                 CleanComment
1    X this service is bad this service is unacceptable
2    Y bad quality product unacceptable quality product
3    Z    not bad actually    not unacceptable actually
```

EXERCISE:06

1. **OUTPUT:**

```
> total
[1] 395
> average
[1] 79
> highest
[1] 91
```

2. **OUTPUT:**

```
> min_temp
[1] 29
> max_temp
[1] 34
```

3. **OUTPUT:**

```
> employee$Name
[1] "Rahul"
> employee$Salary
[1] 52000
```

4. **OUTPUT:**

```
> patient$TestResults
[1] 110 135 128
```

5. **OUTPUT:**

```
> students
  RollNo  Name Dept Marks
1     10 Rohit   IT    78
2     20 Divya  CSE    84
3     30 Kiran  ECE    90
```

6. **OUTPUT:**

```
> total_sales
[1] 207000
```

7. **OUTPUT:**

```
> orders
     Item Price
1  Tablet  1500
2 Earbuds   900
3   Cable   650
```

8. **OUTPUT:**

```
> class_results
  RollNo Total
1    201   245
2    202   230
3    203   260
```

9. **OUTPUT:**

```
> transactions
   Date Amount
1 05-02  -1000
2 06-02   2000
3 07-02   -500
```

10. **OUTPUT:**

```
> records
  Subject Marks
1 English    75
2   Maths    82
3 Biology    88
```

EXERCISE:07

OUTPUT:

```
> Product   TotalSales
  Laptop    110000
  Tablet    135000
  Shoes      27500
  Watch      18000
```

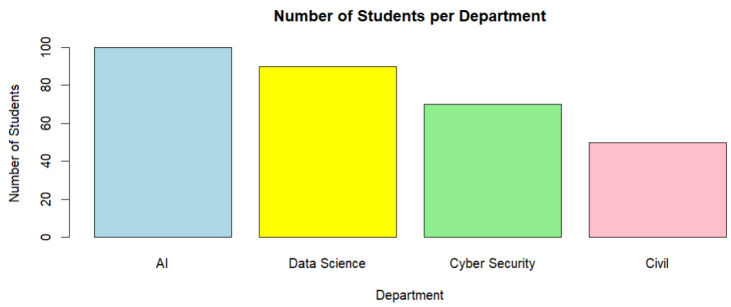
```
> Product   TotalSales
  Tablet    135000
```

```
> Category     Quantity
  Electronics      11
  Fashion          17
```

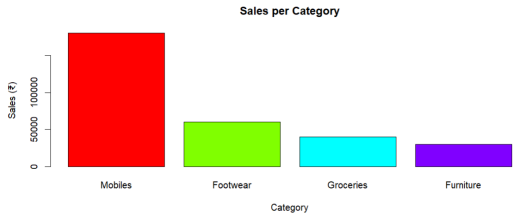
```
> Product   TotalSales
  Laptop    110000
  Tablet    135000
  Shoes      27500
```

EXERCISE:08

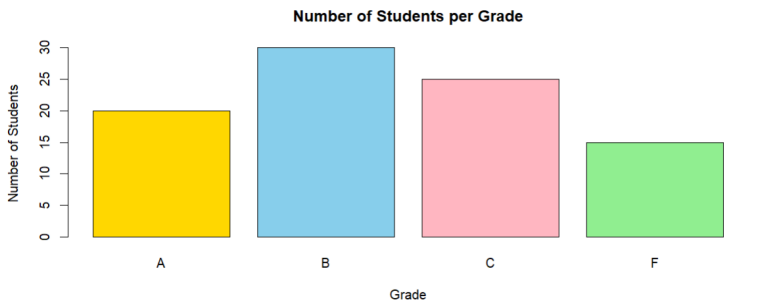
1. **OUTPUT:**



2. **OUTPUT:**

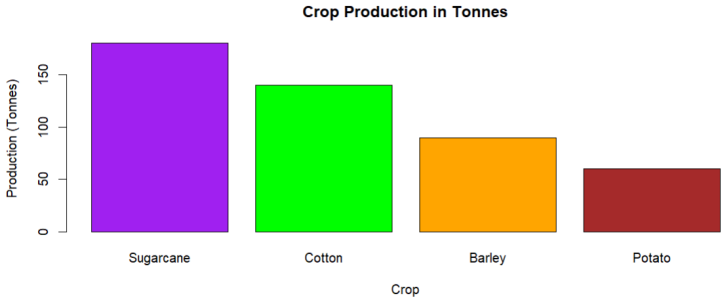


3. **OUTPUT:**



4. **OUTPUT:**

5. **OUTPUT:**



EXERCISE:09

1. **OUTPUT:**

```
> mean_marks
  Maths Science English
   79.8    78.0    81.8
> median_marks
  Maths Science English
   79.0    78.5    81.5
> sd_marks
   Maths  Science  English
9.750214 9.237604 7.238784
> top_student
    Name Maths Science English Total
8 Sowmya    95      93      90   278
> above_avg_math
     Name Maths Science English Total
2   Divya    85      82      78   245
4   Pooja    91      89      94   274
6   Kavya    88      84      86   258
8  Sowmya    95      93      90   278
10  Nisha    80      79      83   242
> Shoes      27500
```

2. **OUTPUT:**

```
> sales_stats
  Product Mean Max Min Variance
1  Tablet  6.4   8   5      1.3
2  Camera  2.4   4   1      1.3
3 Speaker  5.2   6   4      0.7
4   Shoes  9.4  11   8      1.3
5   Watch  3.4   5   2      1.3
6 Printer  7.4   9   6      1.3
7 Monitor  5.4   7   4      1.3
> max_avg_product
  Product Mean Max Min Variance
4   Shoes  9.4  11   8      1.3
> Shoes      27500
```

3. **OUTPUT:**

```
> mean_age
[1] 45.2
> median_age
[1] 46
> range_age
[1] 28 62
> sd_age
[1] 10.62283
> older_patients
   PatientID    Name Age
3        303 Karthik  55
5        305  Ramesh  62
7        307  Sanjay  53
10       310   Kumar  51
> Shoes      27500
```

4. **OUTPUT:**

```
> dept_stats
     Dept Salary.Mean Salary.Median
1 Finance    44666.67         45000
2      HR    39000.00         39000
3      IT    35500.00         35500
> high_earners
  EmpID  Name    Dept Salary
3   203 Kiran Finance  42000
6   206 Deepa Finance  45000
8   208  Ravi      HR  40000
9   209 Sneha Finance  47000
> Shoes      27500
```

5. **OUTPUT:**

```
> mean_rating
[1] 4.2
> median_rating
[1] 4
> mode_rating
[1] 5
> var_rating
[1] 0.6222222
> top_rated
   ProductID Product Rating
1         11      TV      5
4         14   Mixer      5
8         18  Cooler      5
10        20  Vacuum      5
> Shoes      27500
```